

模拟电子技术基础

Analog Device

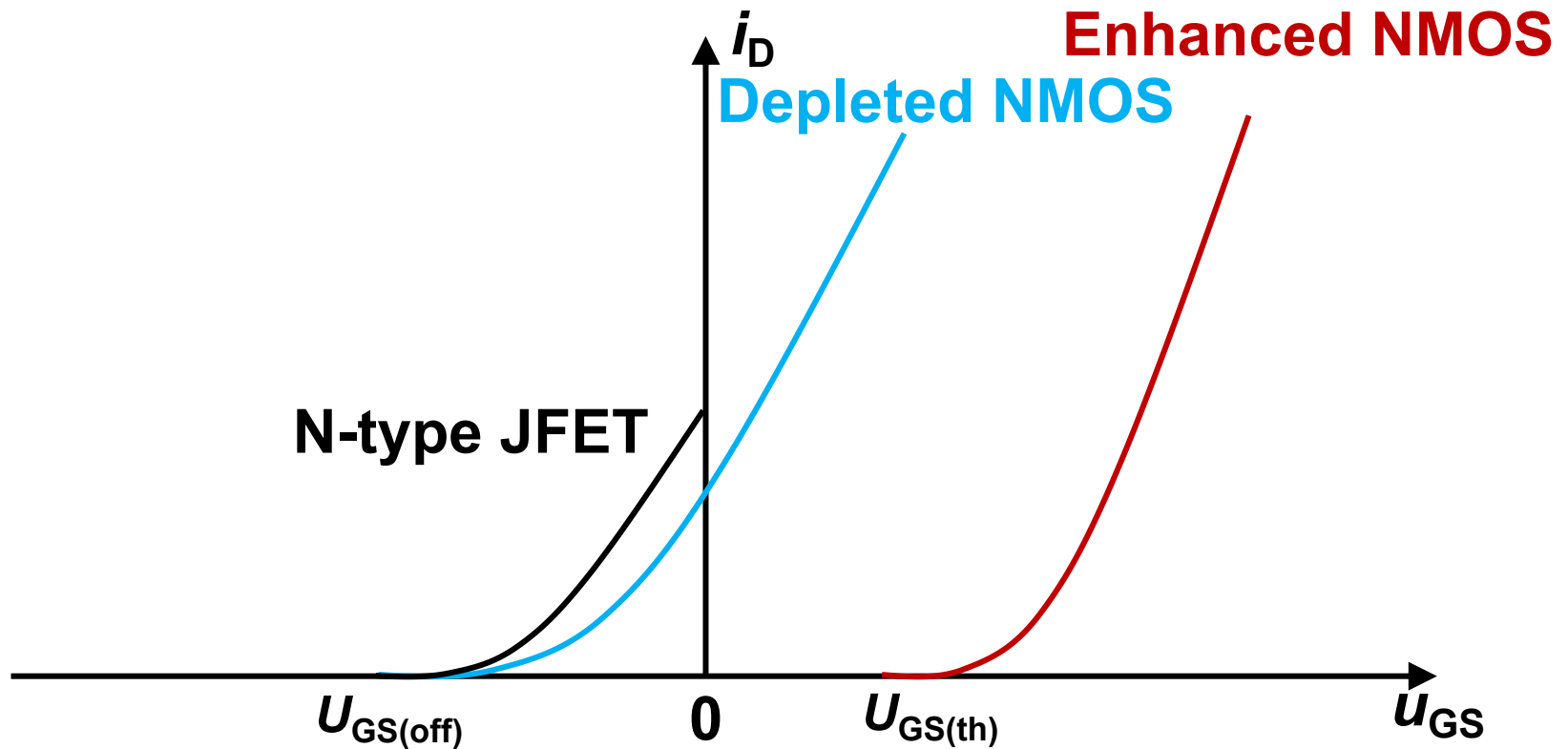
3.4 FET Amplifier Circuit

场效应晶体管放大电路

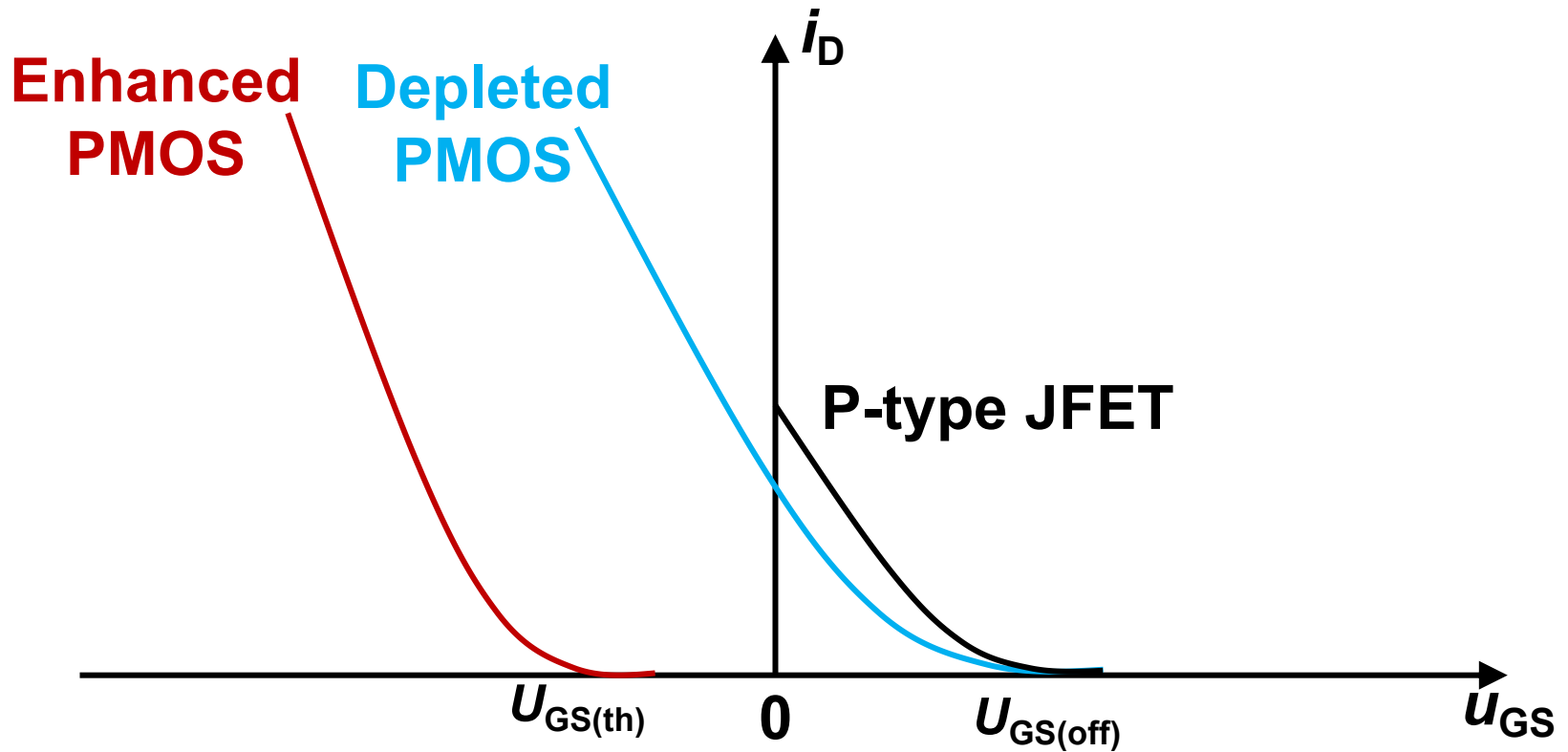
陈晓龙 Chen, Xiaolong

Department of Electrical and Electronic Engineering

Transfer curve for N-type FET 转移特性曲线

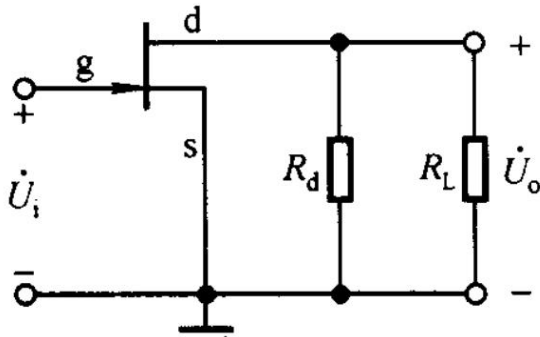


Transfer curve for P-type FET 转移特性曲线

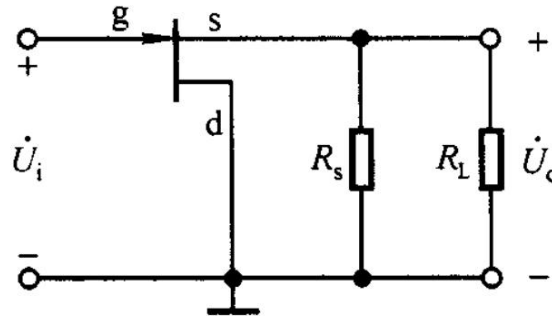


3.4.1 Common-s, -d, -g amplifier

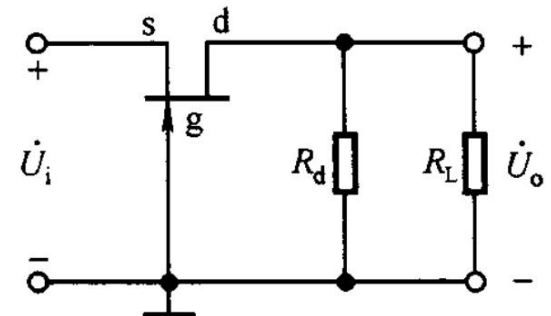
Common-s



Common-d



Common-g



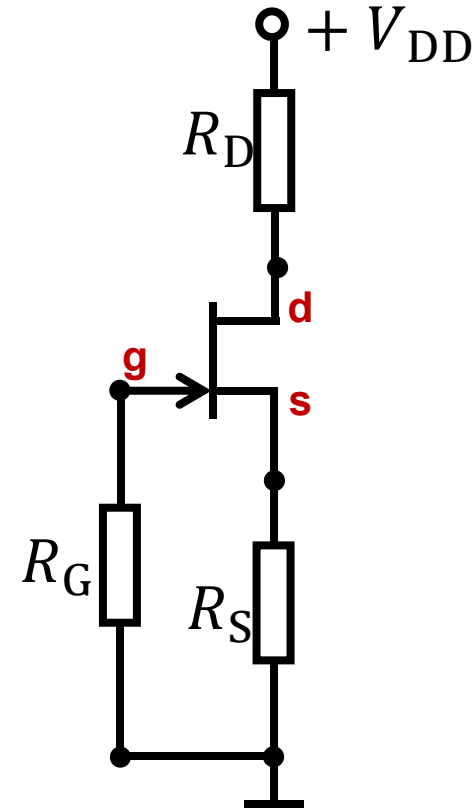
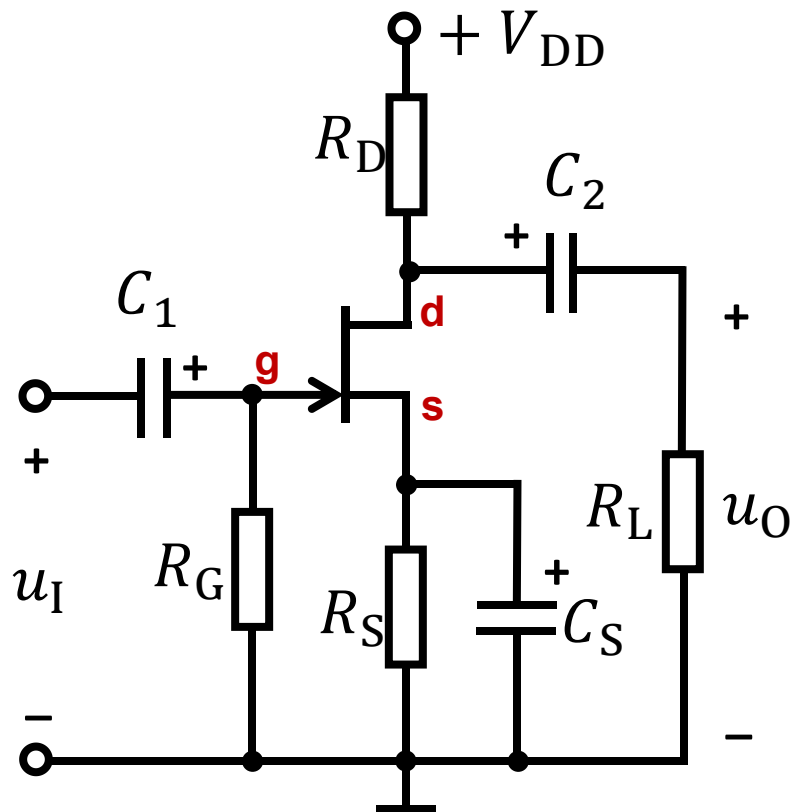
We mainly focused on common-s and -d circuit, since common-g is less used in amplifier circuit

- Self-biased circuit 自给偏压电路
- Voltage divider biasing circuit 分压式偏置电路

3.4.2 Self-biased circuit 自给偏压电路

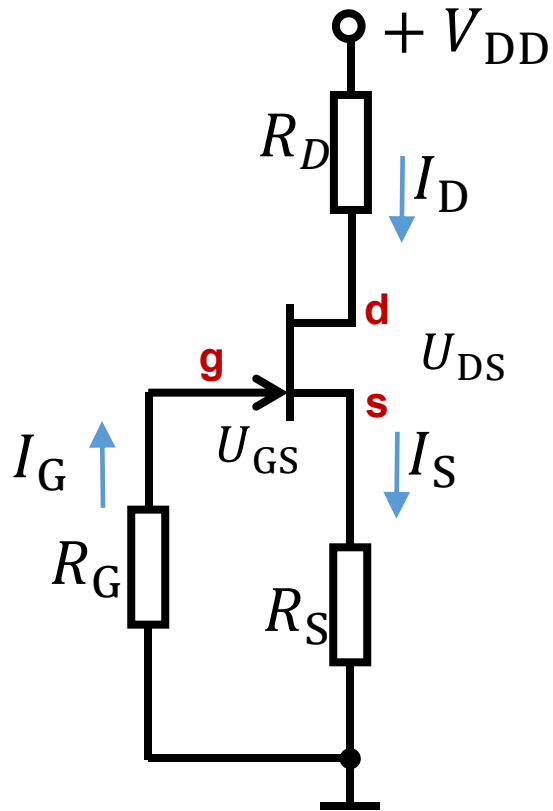
N-type JFET

Analyze DC circuit



No direct voltage supply on gate.

Analyze DC circuit



$$I_G \approx 0 \quad I_D = I_S$$

1) Graphic method 图解法

Input circuit:

$$U_{GS} + I_D R_S = 0$$

$$I_D = -\frac{U_{GS}}{R_S}$$

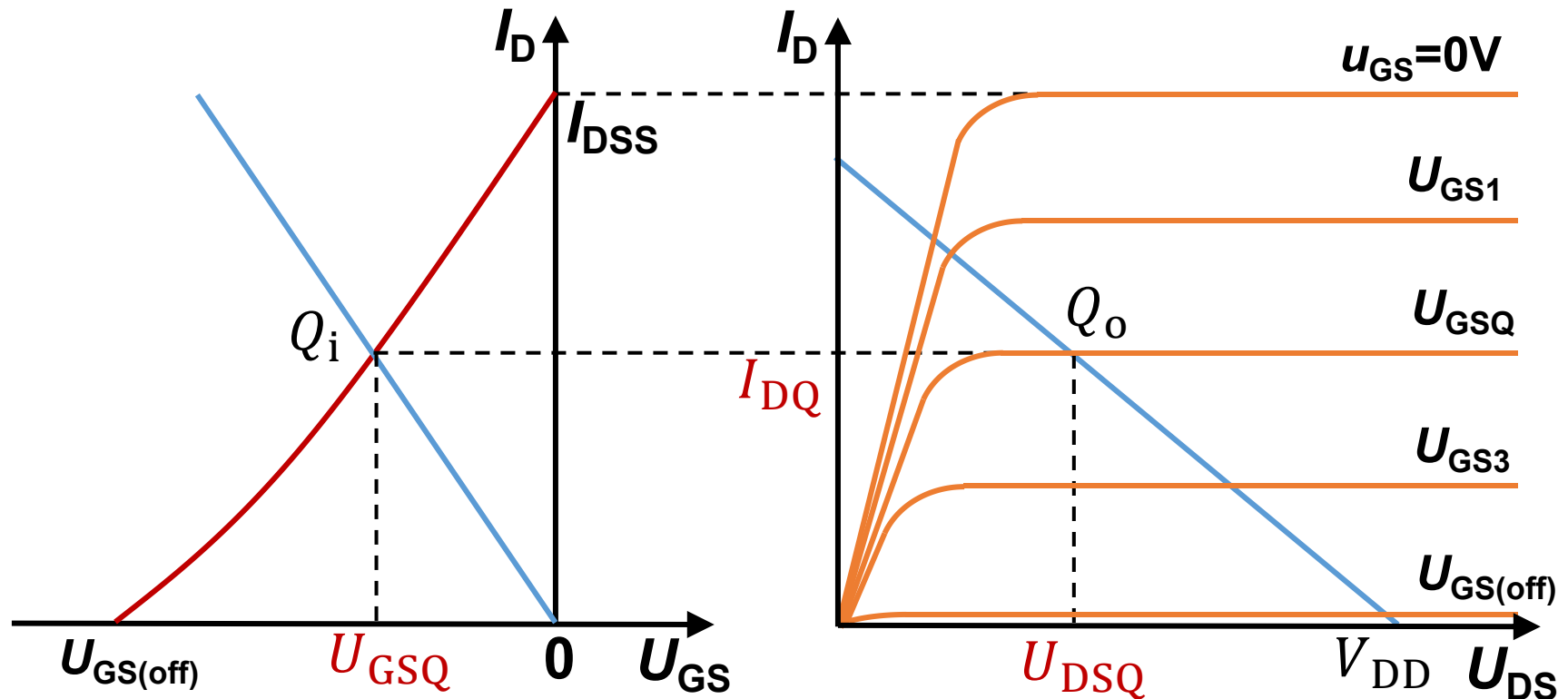
Output circuit:

$$V_{DD} = I_D (R_D + R_S) + U_{DS}$$

$$I_D = \frac{1}{(R_D + R_S)} (V_{DD} - U_{DS})$$

$$I_D = -\frac{U_{GS}}{R_S}$$

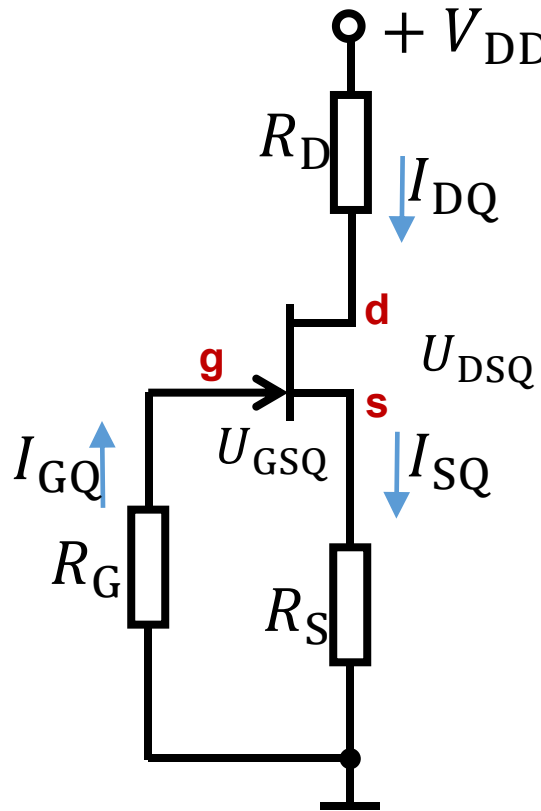
$$I_D = \frac{1}{(R_D + R_S)} (V_{DD} - U_{DS})$$



$U_{GSQ} < 0$: Self-biased circuit is suitable for N-type JFET and depleted NMOSFET

U_{GSQ} , I_{DQ} , I_{SQ} , U_{DSQ} are quiescent working parameters of FET amplifier circuit.

$$I_G \approx 0 \quad I_{DQ} = I_{SQ}$$



2) Approximation method 估算法

Input circuit:

$$\begin{cases} U_{GSQ} + I_{DQ}R_S = 0 \\ I_{DQ} = I_{DSS}\left(1 - \frac{U_{GSQ}}{U_{GS(off)}}\right)^2 \end{cases}$$

➡ I_{DQ}, U_{GSQ}

Output circuit:

$$V_{DD} = I_{DQ}(R_D + R_S) + U_{DSQ}$$

➡ U_{DSQ}



Q1: As shown in the figure, $V_{DD}=18V$, $R_D=3k\Omega$, $R_S=1k\Omega$ and $R_G=1M\Omega$. FET has $I_{DSS}=7mA$ and $U_{GS(off)}=-8V$. Ask: the value of U_{GSQ} , I_{DQ} and U_{DSQ} .

Answer: Input circuit:

$$\begin{cases} U_{GSQ} + I_{DQ}R_S = 0 \\ I_{DQ} = I_{DSS}\left(1 - \frac{U_{GSQ}}{U_{GS(off)}}\right)^2 \end{cases} \quad \begin{cases} U_{GSQ} + I_{DQ} = 0 \\ I_{DQ} = 7 \times \left(1 + \frac{U_{GSQ}}{8}\right)^2 \end{cases}$$

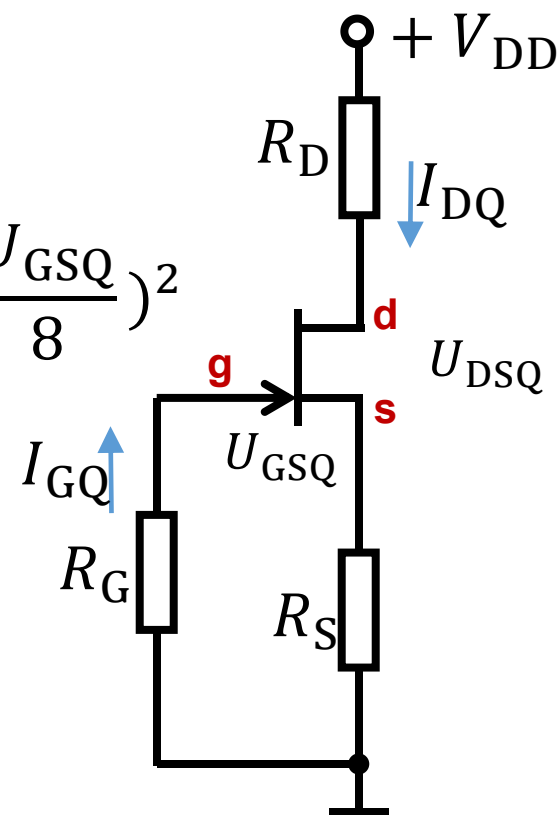
$$I_{DQ} = 2.9 \text{ mA} \quad U_{GSQ} = -2.9 \text{ V}$$

Output circuit:

$$V_{DD} = I_{DQ}(R_D + R_S) + U_{DSQ}$$

$$18 = 2.9 \times (3 + 1) + U_{DSQ}$$

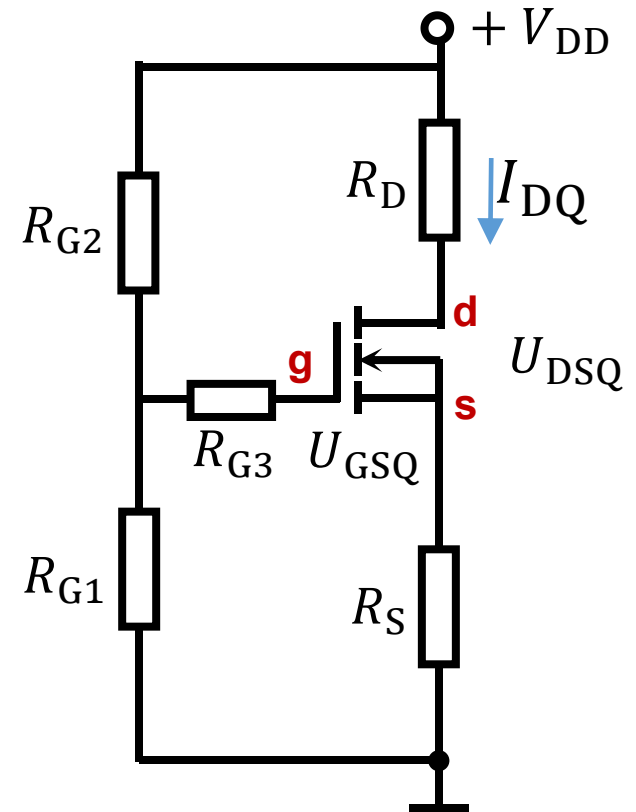
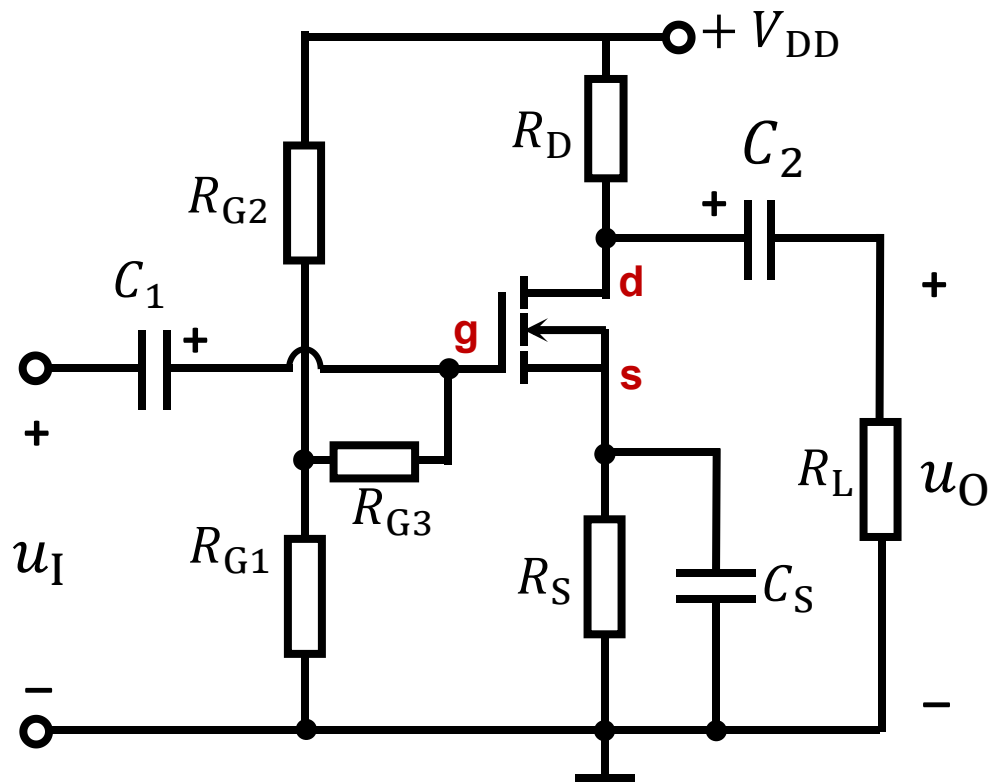
$$U_{DSQ} = 6.4 \text{ V}$$



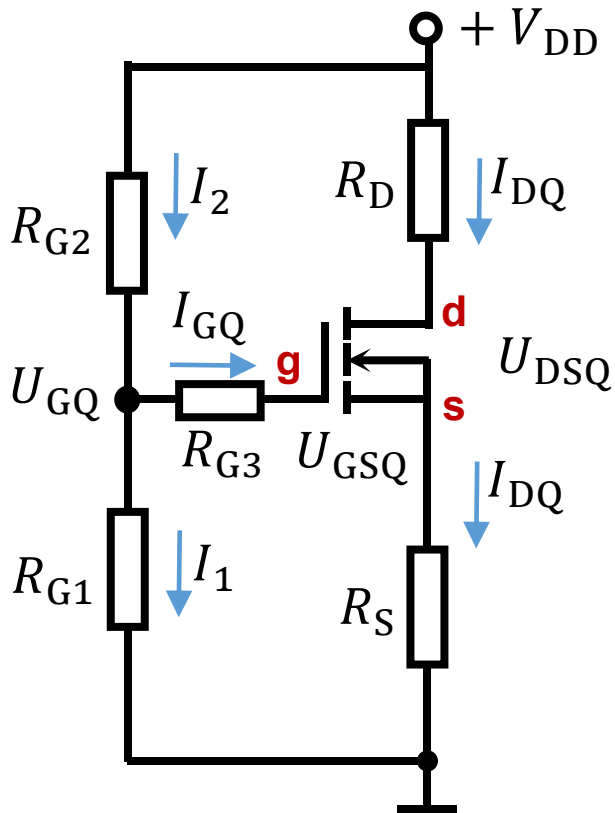
3.4.3 Voltage divider biasing circuit

Enhanced N-type MOS

Analyze DC circuit



Analyze DC circuit



$$I_{GQ} = 0 \quad I_1 = I_2$$

$$U_{GQ} = \frac{R_{G1}}{R_{G1} + R_{G2}} \cdot V_{DD}$$

$$U_{SQ} = I_{DQ} \cdot R_S$$

$$\left\{ \begin{aligned} U_{GSQ} &= \frac{R_{G1}}{R_{G1} + R_{G2}} \cdot V_{DD} - I_{DQ} \cdot R_S \\ I_{DQ} &= I_{DO} \left(\frac{U_{GSQ}}{U_{GS(th)}} - 1 \right)^2 \end{aligned} \right.$$

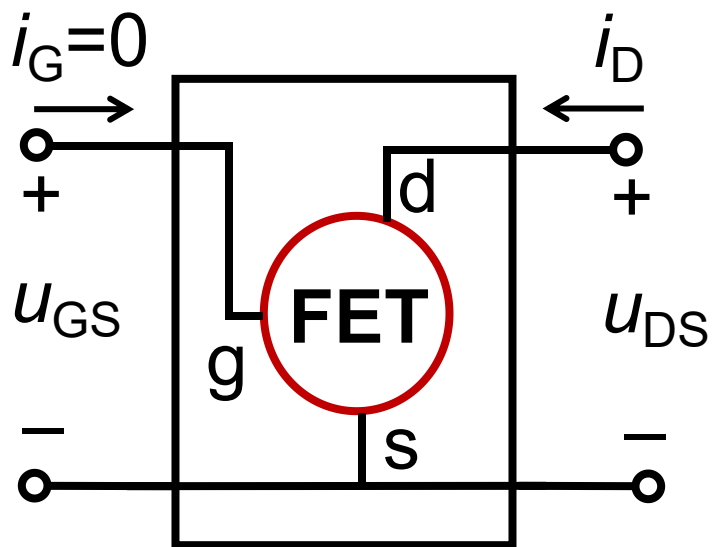
$$U_{DSQ} = V_{DD} - I_{DQ}(R_D + R_S)$$

$U_{GSQ} > 0$ and $U_{GSQ} < 0$ can both be achieved: Voltage divider biasing circuit is suitable for JFET, depleted and enhanced MOSFET

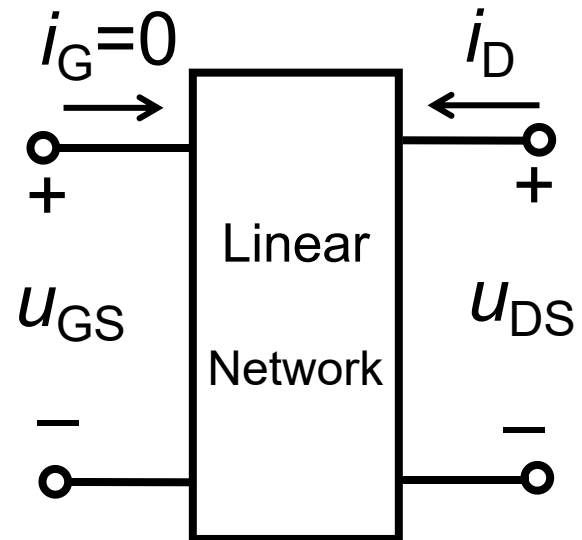
3.4.4 Dynamic analysis of FET amplifier circuit

Find the equivalent circuit of FET: to replace FET with a linear network

Transistor



Equivalent circuit



$$① \quad i_D = f(u_{GS}, u_{DS}) \quad u_{GS} = u_{GS}$$

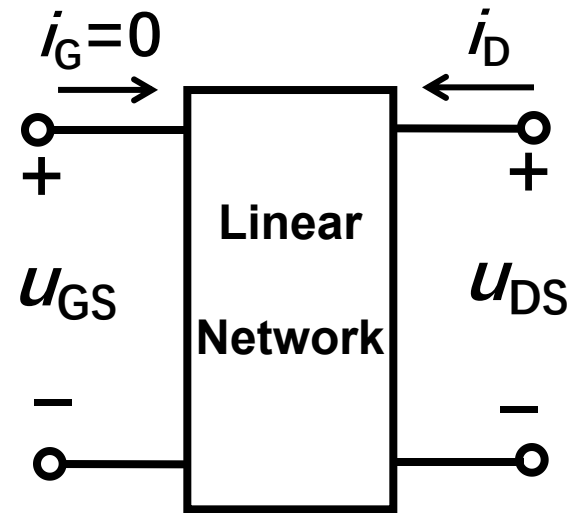
$$② \quad di_D = \frac{\partial i_D}{\partial u_{GS}} \Big|_{U_{DSQ}} du_{GS} + \frac{\partial i_D}{\partial u_{DS}} \Big|_{U_{GSQ}} du_{DS}$$

Transconductance g_m 互导/跨导

$$g_m = \frac{di_D}{du_{GS}} \Big|_{U_{DSQ}}$$

$$g_{ds} = \frac{1}{r_{ds}} = \frac{di_D}{du_{DS}} \Big|_{U_{GSQ}}$$

$$di_D = g_m du_{GS} + \frac{1}{r_{ds}} du_{DS}$$

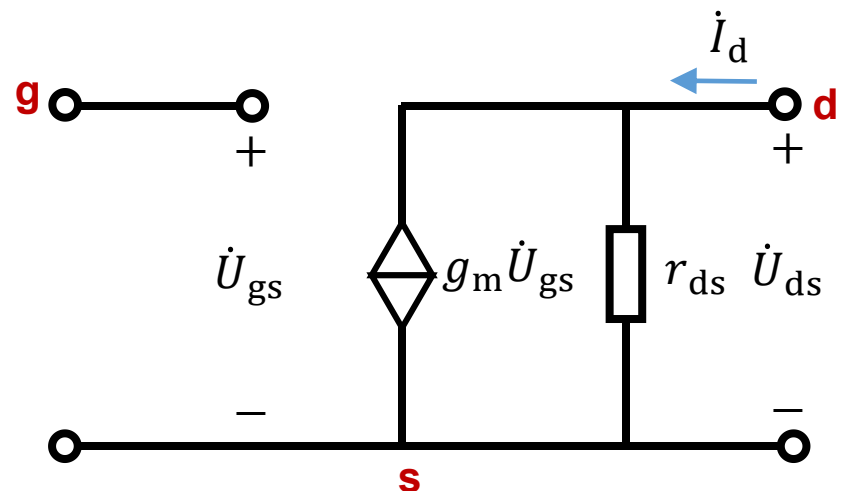
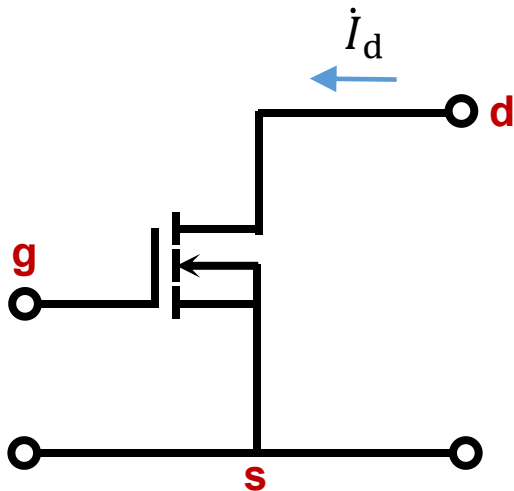


$$di_D = g_m du_{GS} + \frac{1}{r_{ds}} du_{DS}$$

Input signal i_D is small 小信号条件

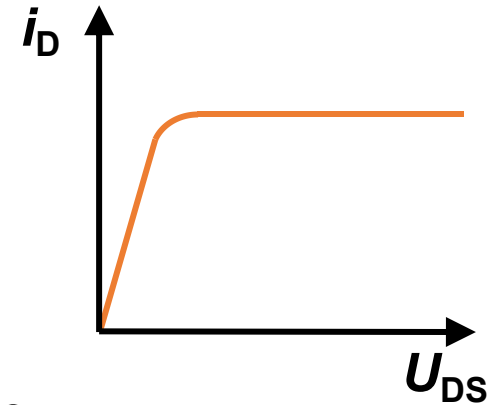
di_D can be replaced by a small quantity: i_d or $\dot{I}_d$

$$\dot{U}_{gs} = \dot{U}_{gs} \quad \dot{I}_d = g_m \dot{U}_{gs} + \frac{1}{r_{ds}} \dot{U}_{ds}$$



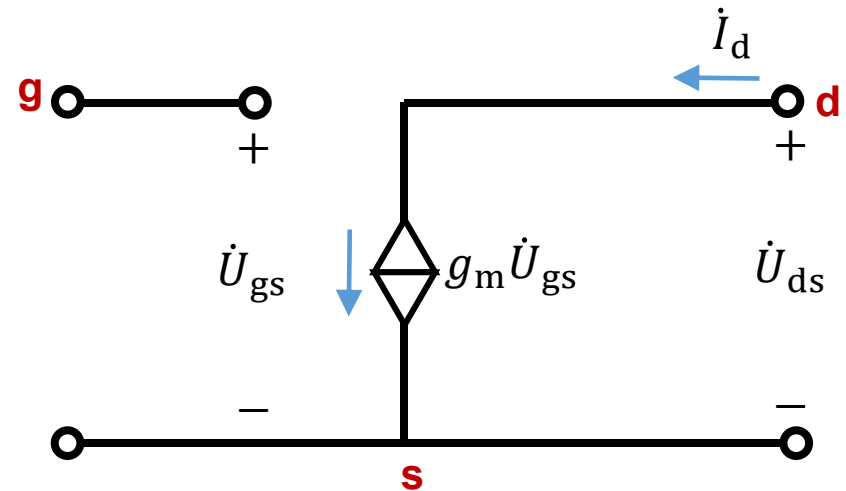
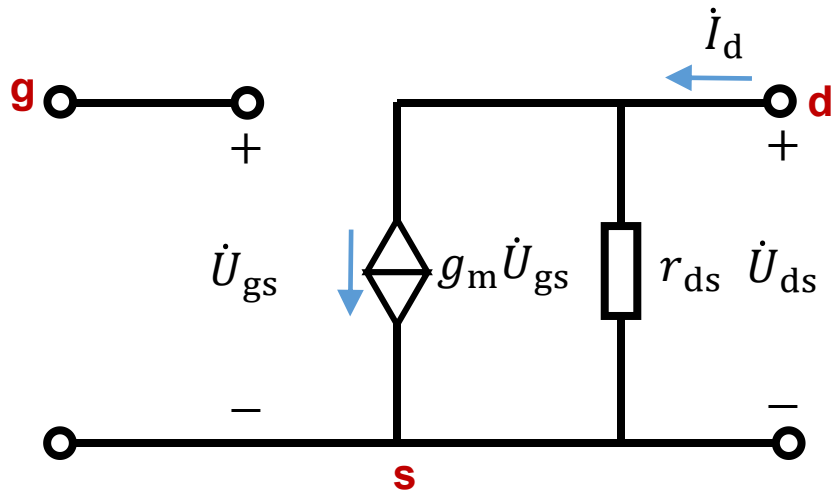
$$\dot{i}_d = g_m \dot{U}_{gs} + \frac{1}{r_{ds}} \dot{U}_{ds}$$

$$g_{ds} = \frac{1}{r_{ds}} = \left. \frac{di_D}{du_{DS}} \right|_{U_{GSQ}}$$

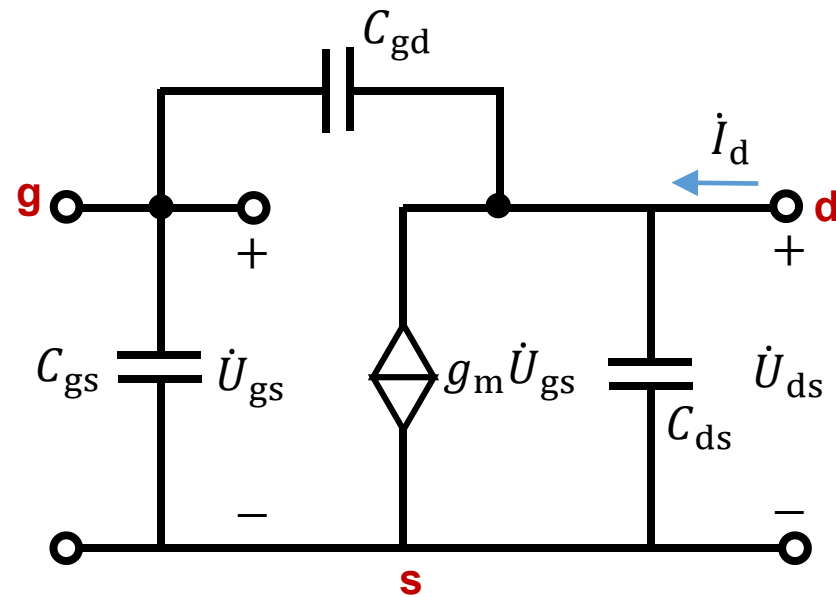


In constant current region, $\left. \frac{di_D}{du_{DS}} \right|_{U_{GSQ}} \approx 0, r_{ds} \rightarrow \infty$.

Simplified model

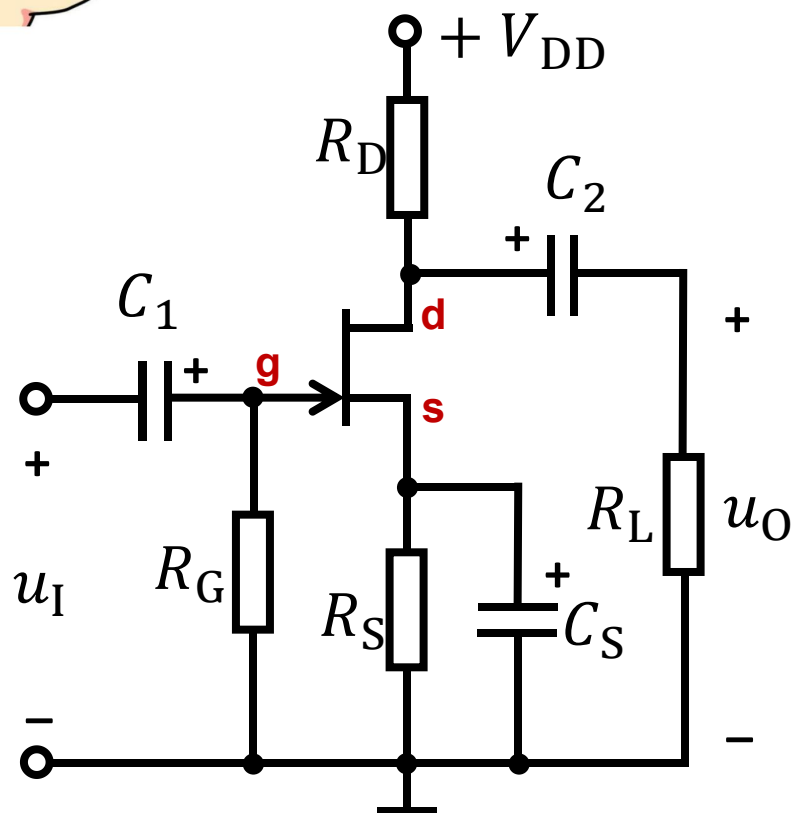


High frequency model





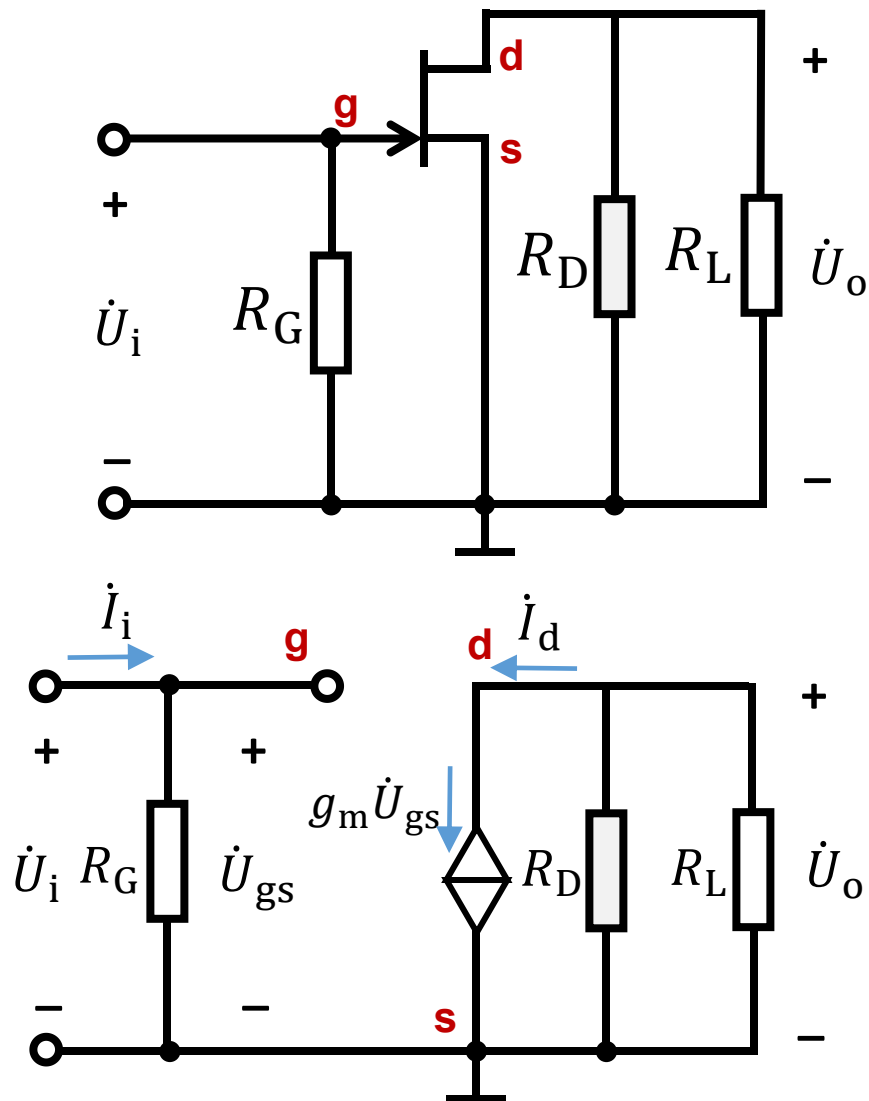
Common-s amplifier circuit

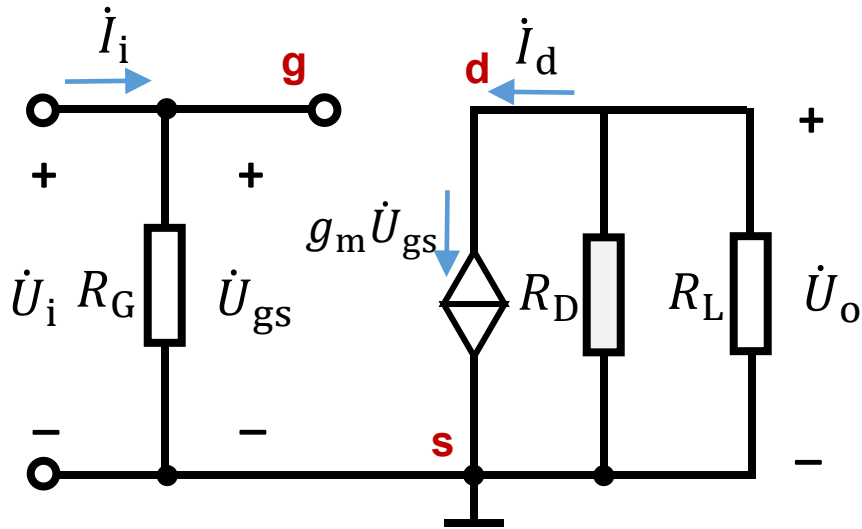


N-type JFET

Self-biased circuit

AC circuit





Voltage gain: $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$

$$\dot{U}_i = \dot{U}_{gs}$$

$$\dot{U}_o = -\dot{I}_d (R_D // R_L)$$

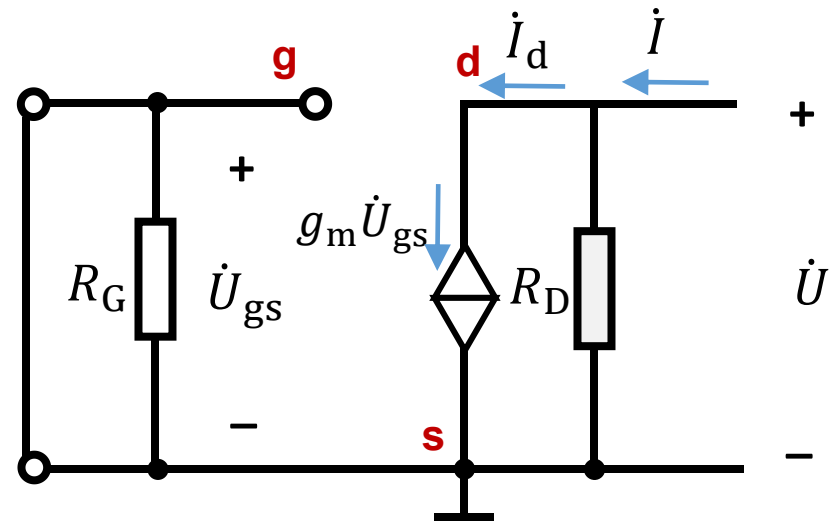
$$= -g_m \dot{U}_{gs} (R_D // R_L)$$

$$\dot{A}_u = -g_m (R_D // R_L)$$

Input resistance:

$$R_i = \frac{\dot{U}_i}{\dot{I}_i} \quad R_i = R_G$$

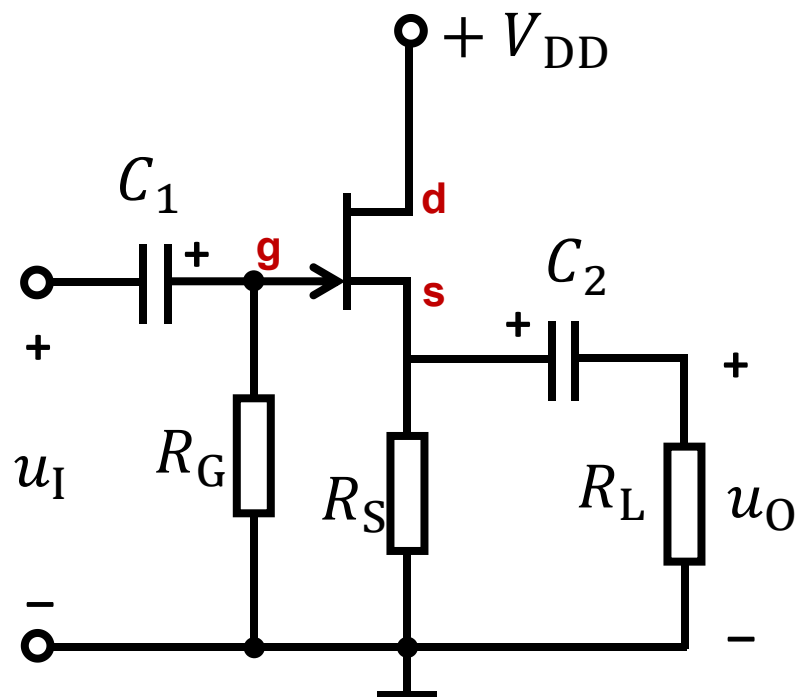
Output resistance:



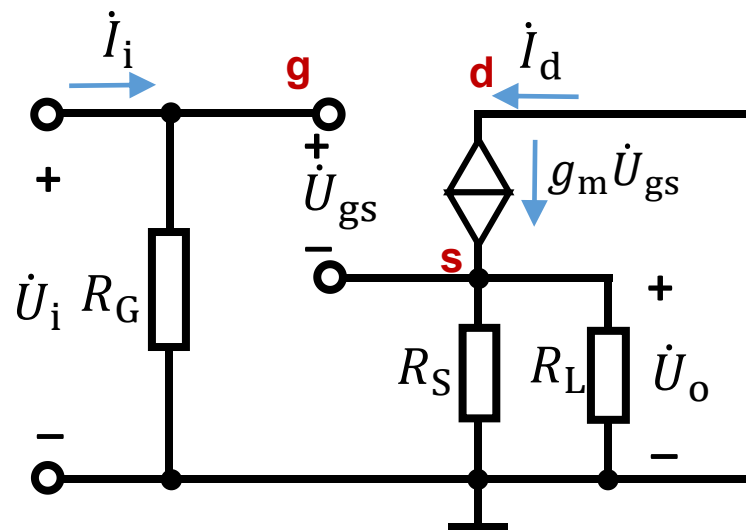
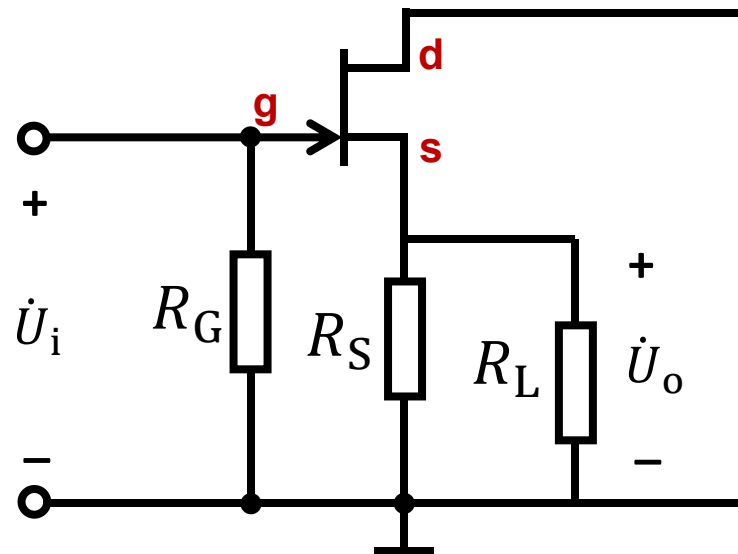
$$\dot{U}_{gs} = 0 \quad R_o = R_D$$

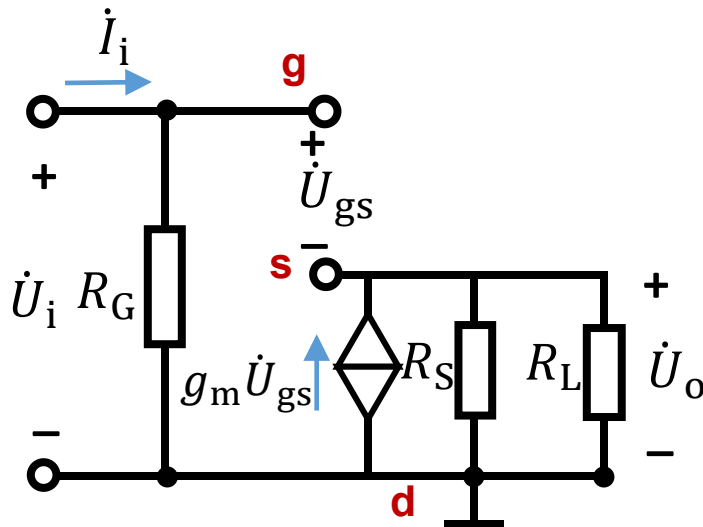
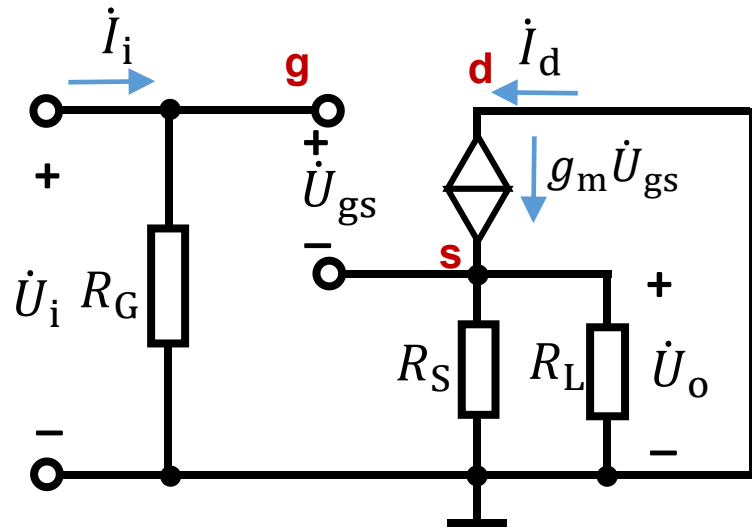
Common-d amplifier circuit

AC circuit



N-type JFET
Self-biased circuit





Voltage gain: $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$

$$\dot{U}_i = \dot{U}_{gs} + \dot{U}_o$$

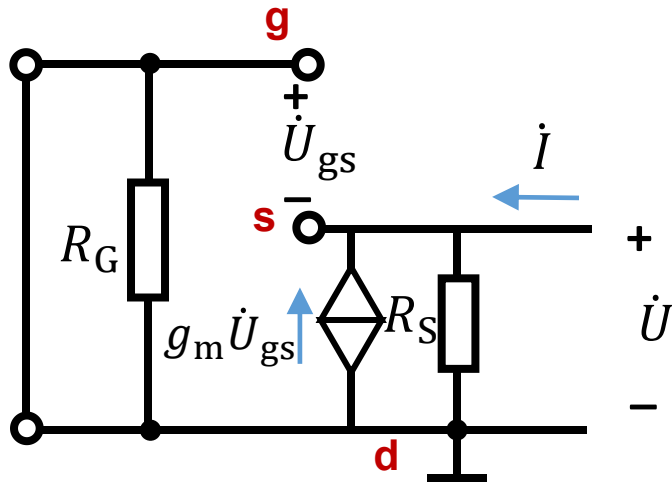
$$\dot{U}_o = g_m \dot{U}_{gs} (R_S // R_L)$$

$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_{gs} + \dot{U}_o}$$

$$= \frac{g_m (R_S // R_L)}{1 + g_m (R_S // R_L)}$$

Input resistance:

$$R_i = R_G$$



Output resistance:

$$R_o = \frac{\dot{U}}{\dot{I}}$$

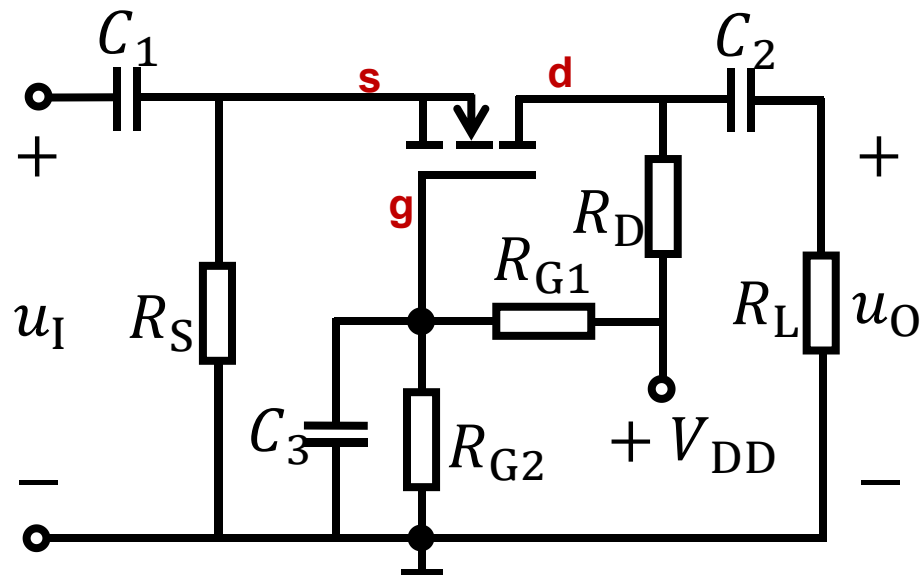
$$\dot{U}_{gs} + \dot{U} = 0$$

$$\dot{U} = (\dot{I} + g_m \dot{U}_{gs}) R_s$$

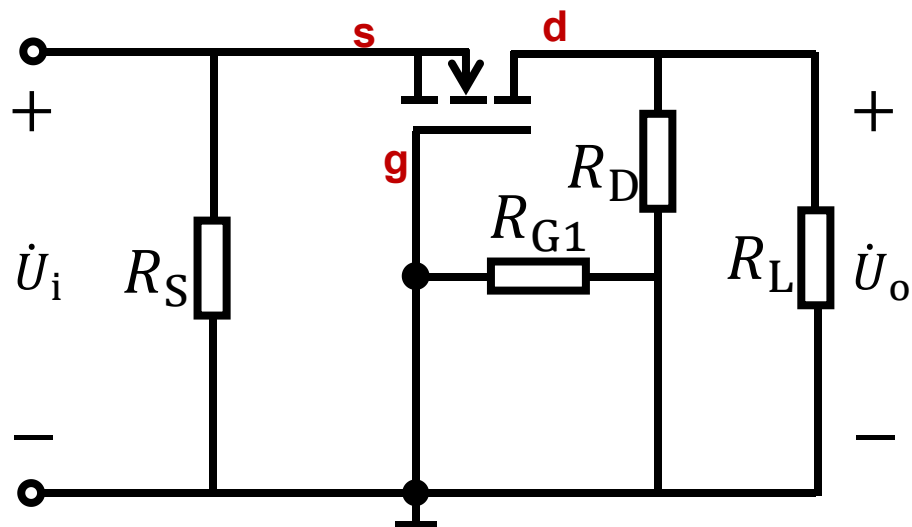
$$\dot{I} = -\frac{\dot{U}_{gs}}{R_s} - g_m \dot{U}_{gs}$$

$$R_o = \frac{1}{\frac{1}{R_s} + g_m} = R_s // \frac{1}{g_m}$$

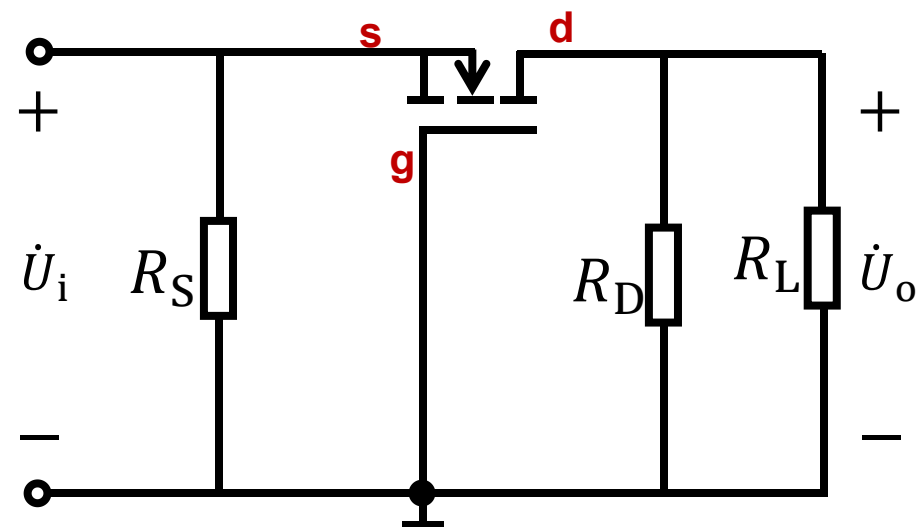
Common-g amplifier circuit

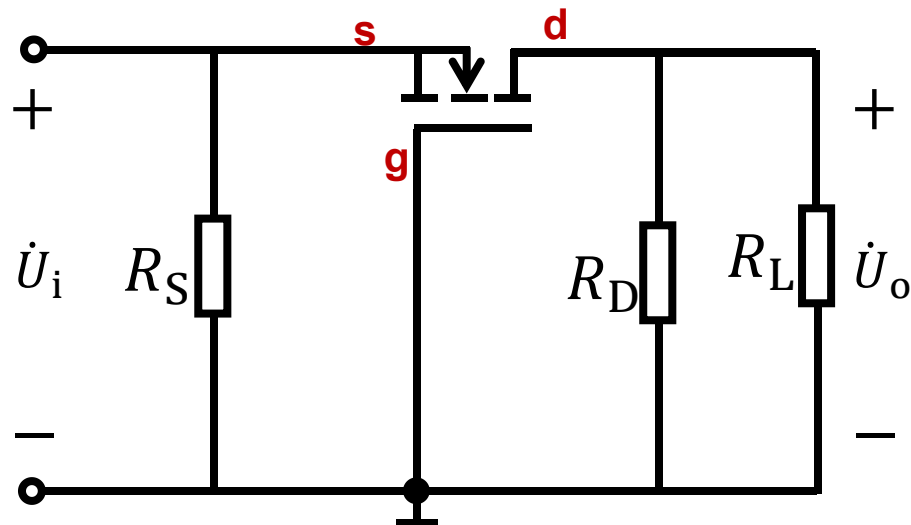


AC circuit



Enhanced N-type MOSFET Voltage divider biasing circuit



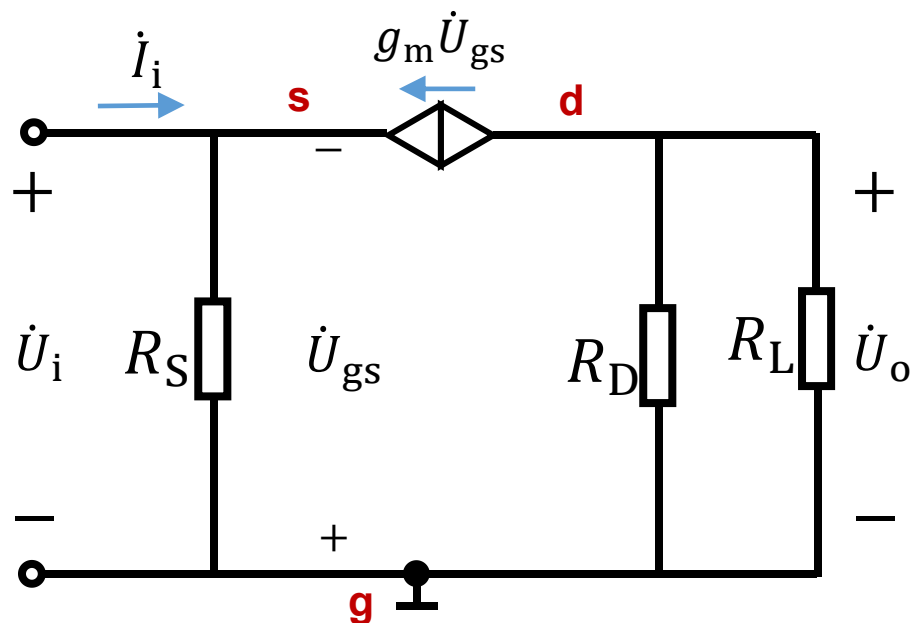


Voltage gain: $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$

$$\dot{U}_i = -\dot{U}_{gs}$$

$$\dot{U}_o = -g_m \dot{U}_{gs} (R_D // R_L)$$

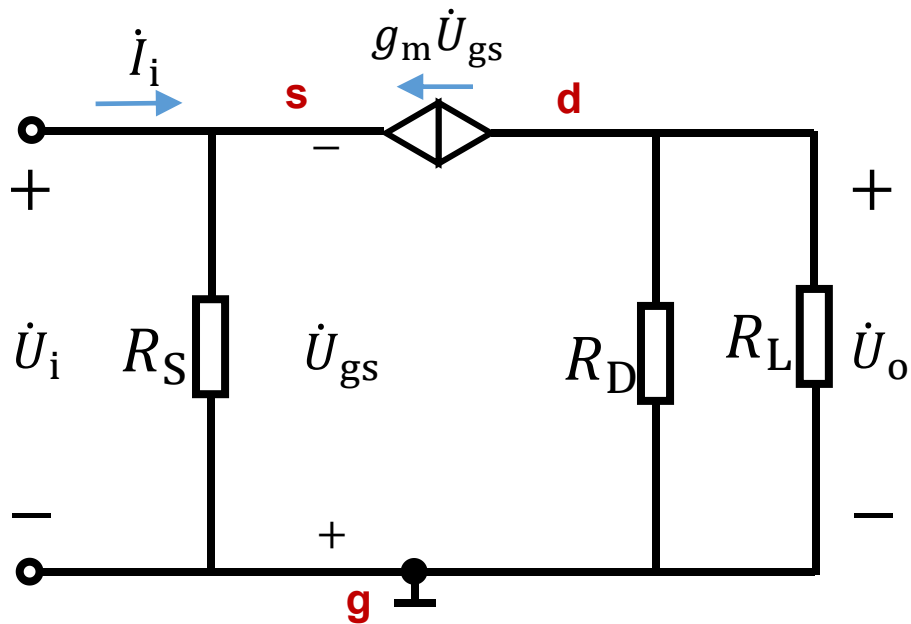
$$\dot{A}_u = g_m (R_D // R_L)$$



Input resistance: $R_i = \frac{\dot{U}_i}{\dot{I}_i}$

$$-\dot{U}_{gs} = (\dot{I}_i + g_m \dot{U}_{gs}) R_S$$

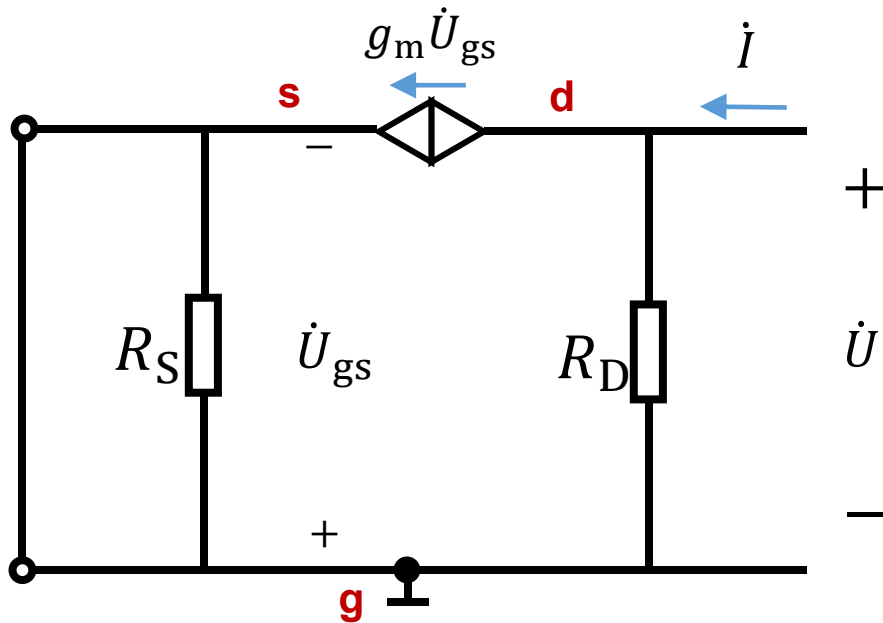
$$R_i = \frac{1}{\frac{1}{R_S} + g_m} = R_S // \frac{1}{g_m}$$



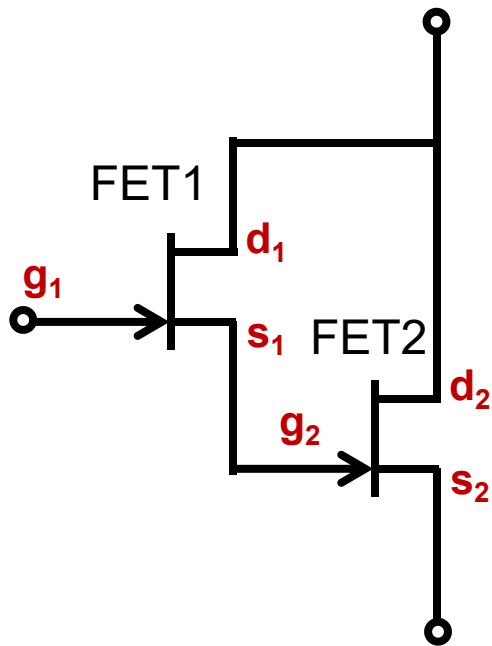
Output resistance:

$$R_o = \frac{\dot{U}}{\dot{I}} \bigg|_{R_L=\infty, \dot{U}_i=0}$$

$$R_o = R_D$$



3.5 Composite transistor using FET

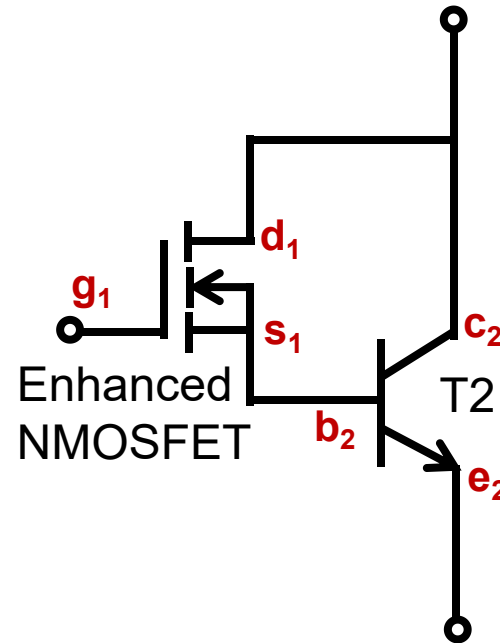
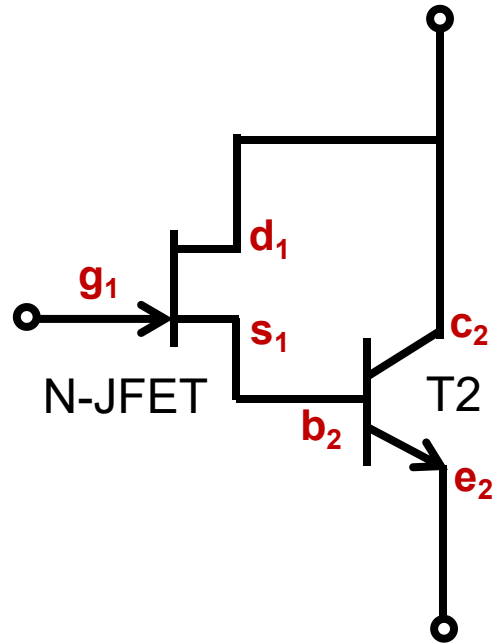


Q: Can the left configuration amplify signals?

Ans: No.

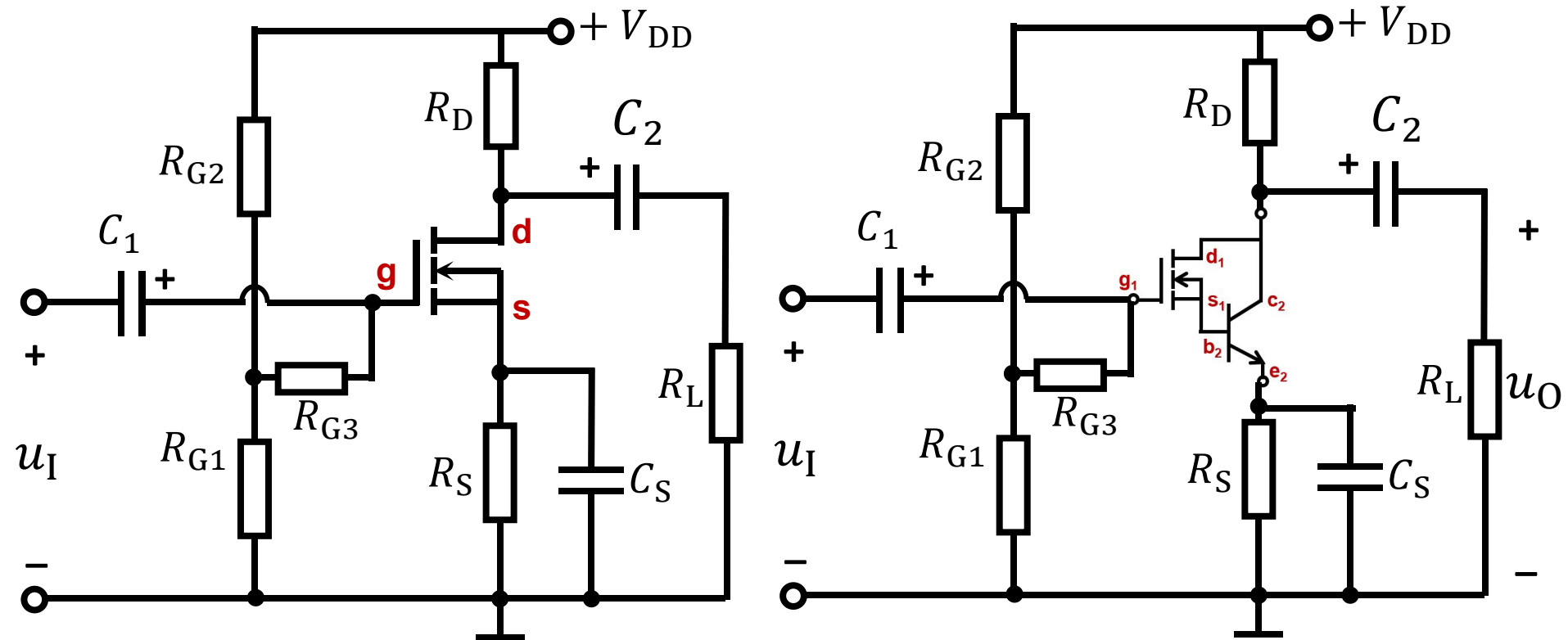
For FETs, current through gate is very small $I_g \approx 0$. So we cannot connect s_1 or d_1 to g_2 .

We can connect FET with BJT together to form a composite transistor



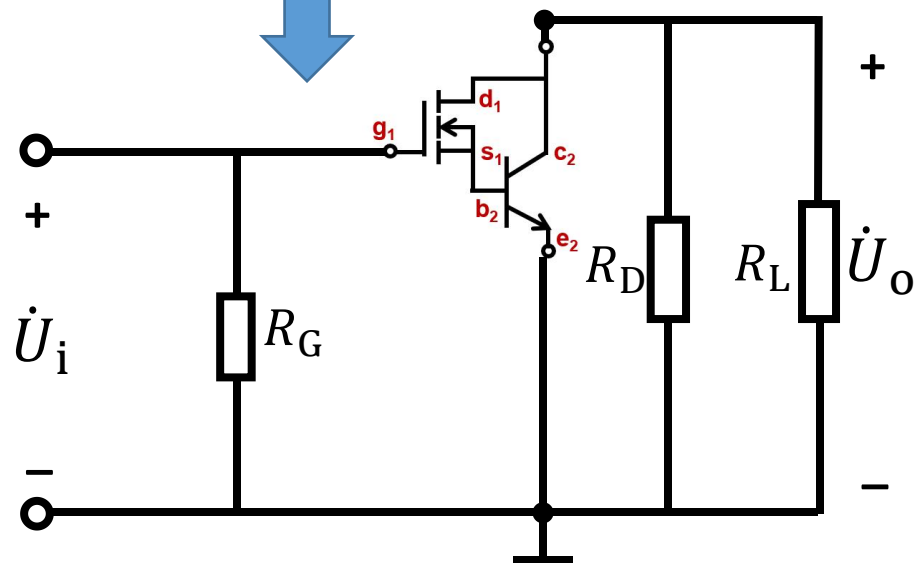
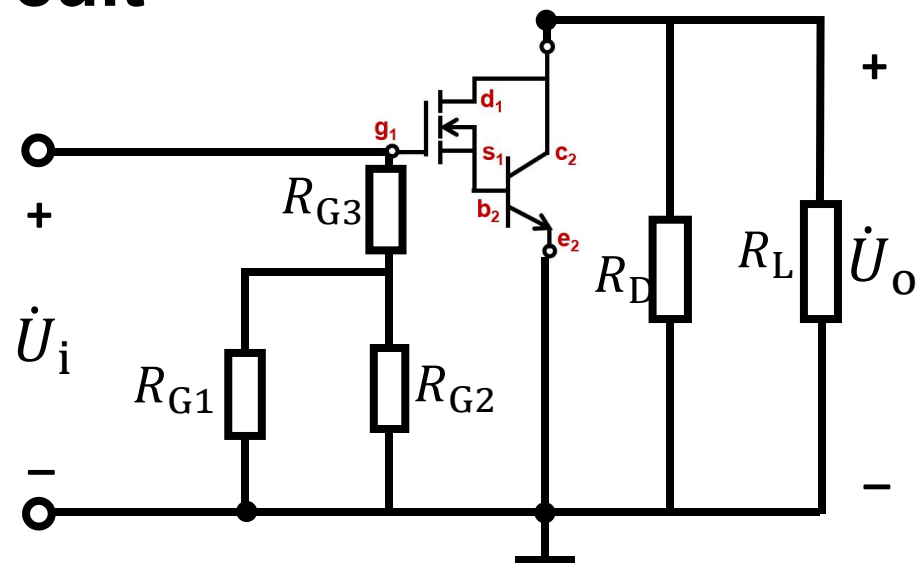
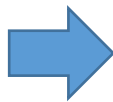
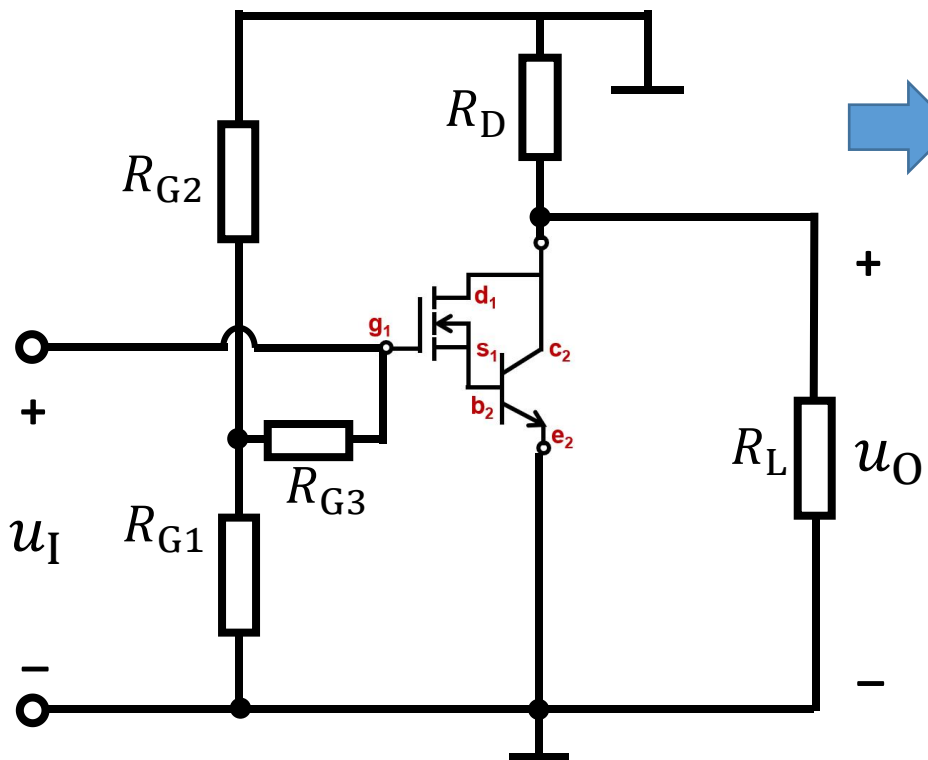
The type of a composite transistor is determined by the type of the first transistor

Example: the AC voltage gain of right figure?



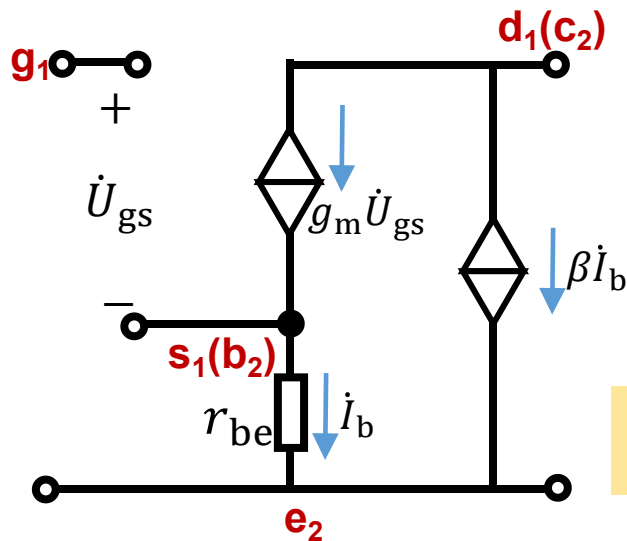
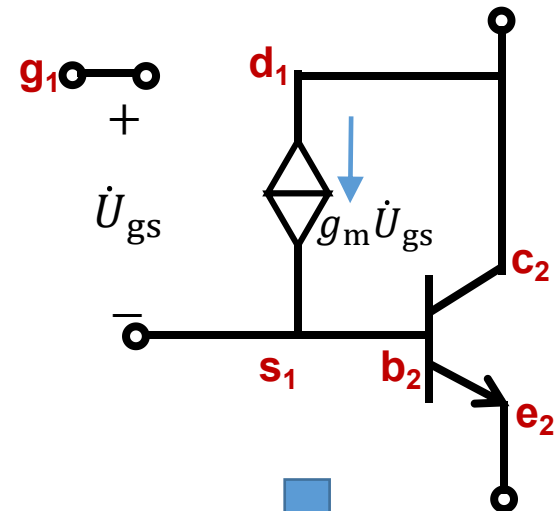
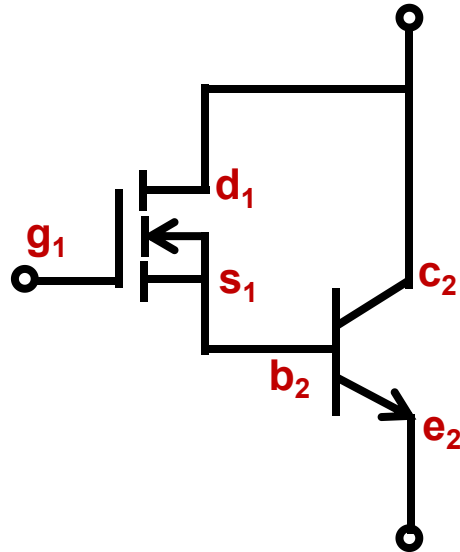
Analyze the DC circuit by yourself.

AC circuit

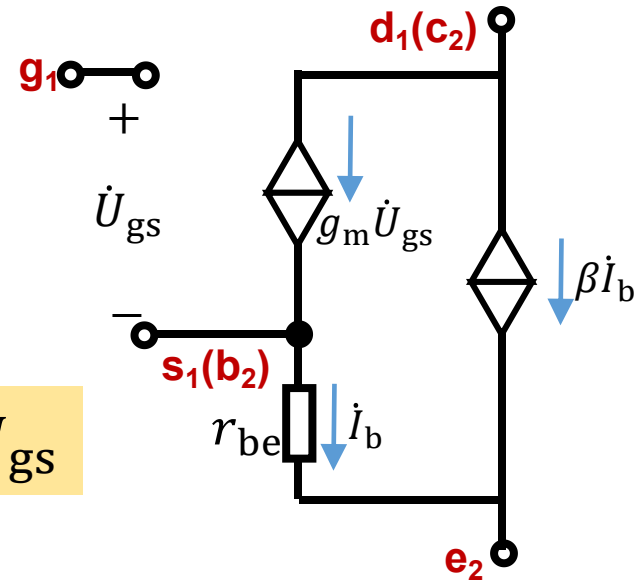
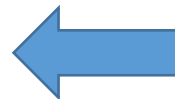


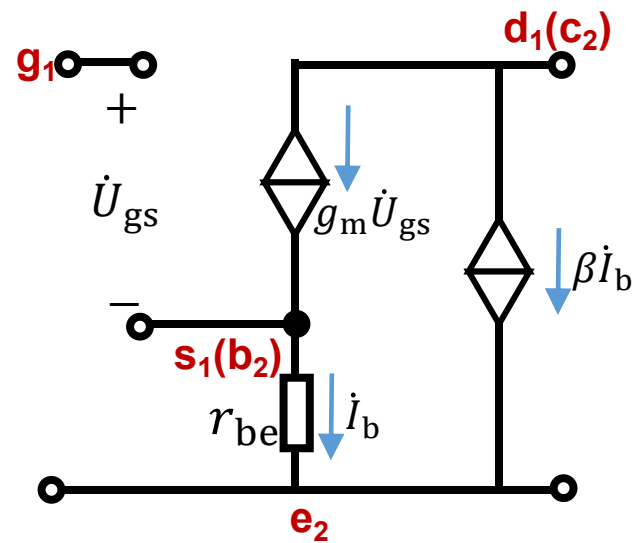
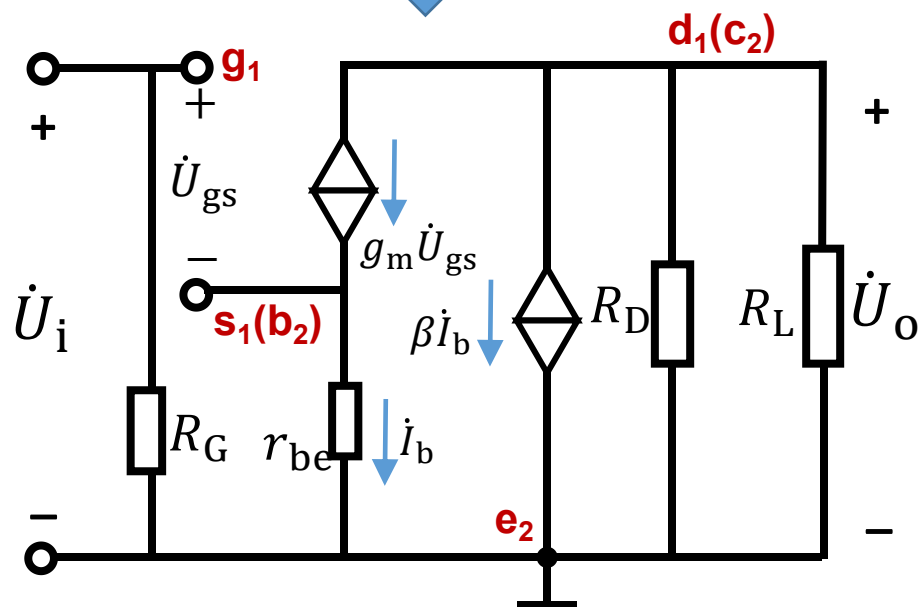
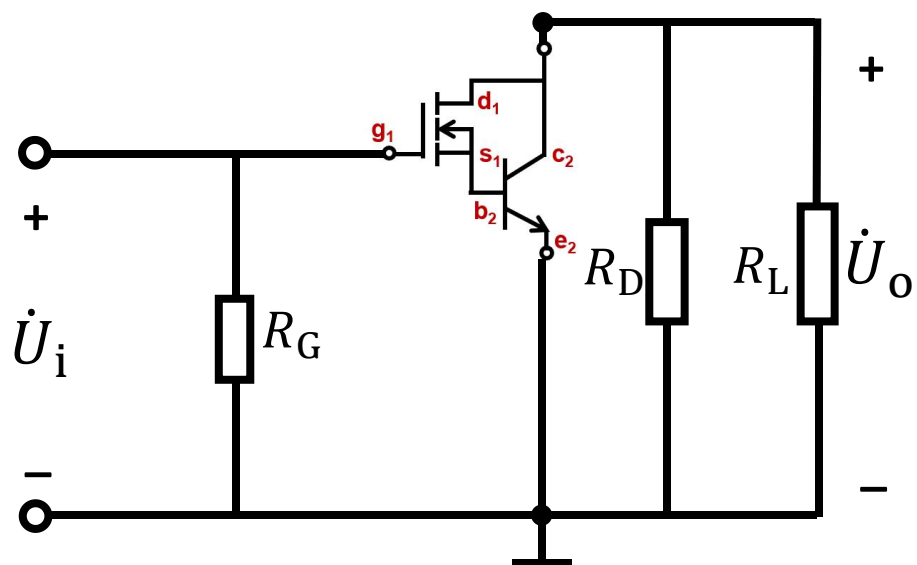
$$R_G = R_{G3} + R_{G1} || R_{G2}$$

AC equivalent circuit of the composite transistor

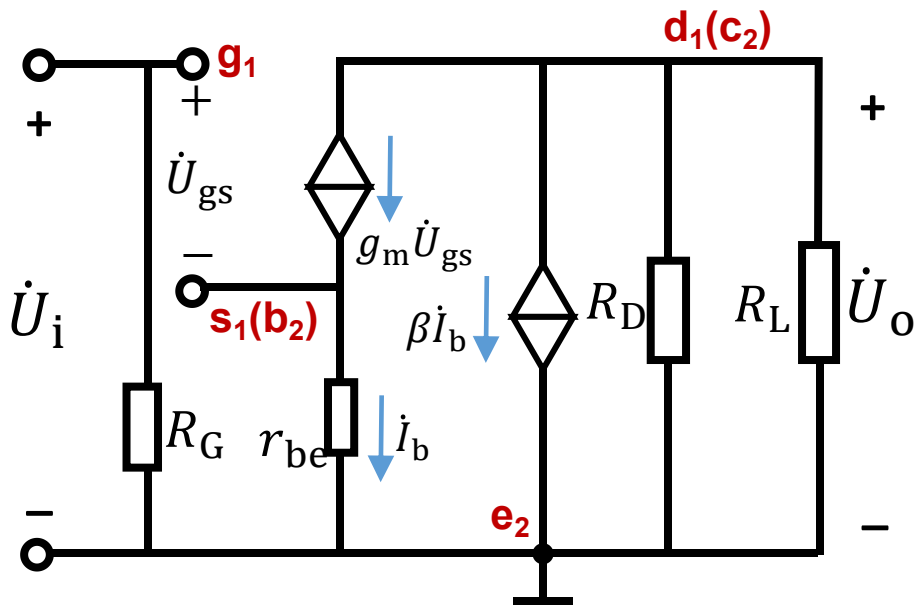


$$i_b = g_m \dot{U}_{gs}$$





$$\dot{I}_b = g_m \dot{U}_{gs}$$



Voltage gain: $\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i}$

$$\begin{aligned}\dot{U}_i &= \dot{U}_{gs} + \dot{I}_b r_{be} \\ &= \dot{U}_{gs} (1 + g_m r_{be})\end{aligned}$$

$$\begin{aligned}\dot{U}_o &= - (g_m \dot{U}_{gs} + \beta \dot{I}_b) (R_D // R_L) \\ &= - (1 + \beta) g_m \dot{U}_{gs} (R_D // R_L)\end{aligned}$$

$$\dot{A}_u = - \frac{(1 + \beta) g_m (R_D // R_L)}{1 + g_m r_{be}}$$

Calculate the input resistance and output resistance by yourself.

Homework 3-1: refer to blackboard

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